An Adaptive Frequency Sampling Method for Frequency Selective Surface design Exploiting Steor-Bulirsch Algorithm

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Abstract—In this paper, the adaptive frequency sampling (AFS) method with steor-bulirsch (SB) algorithm is proposed in conjunction with the method of moments (MOM) to calculate the broadband electromagnetic scattering performance of the frequency selective surface (FSS). Comparing with other sampling methods, the Store-Bulirsch algorithm is a recursive tabular method and requires no matrix inversion, so it can obtain a rational interpolation function without suffering from singularity problems. Furthermore, since we use the AFS strategy to interpolate the final results, no more information is needed except the sampled frequency *f* and its true value, so it can be widely used in most of the electromagnetic (EM) simulation. Numerical results show that our sampling strategy has a good performance in both CPU time saving and computation accuracy.

Keywords-**Adaptive sampling; Store-Bulirsch algorithm; Method of moments (MOM); Frequency selective surface (FSS); Electromagnetic scattering**

I INTRODUCTION

Frequency selective surfaces (FSS) are widely used as subreflectors in many dual reflector systems in order to provide multi-frequency capabilities. When the structure of FSS is large relative to wavelengths of interest, the large number of required frequency-domain computations may require significant computational resources. One effective method that alleviates the computation time is to interpolate the required data by analytic model-based function, among which the rational function interpolation proves to be the most versatile, effective and apposite one. In this paper, we investigate a new rational function interpolation method based on Store-Bulirsch algorithm [1-3] (So called SB-AFS hereafter), which hybridizes with the method of moments (MOM) to accelerate the reflective and perspective parameter computation of the frequency selective surface (FSS) [4-6].

We address reducing computational costs by performing numerical calculations only at select frequencies, with a parametric model employed to fill for results at the remaining frequencies. Related ideas have been considered by other authors, for example, the asymptotic waveform evaluation (AWE) method [7] and the model-based parameter estimation (MBPE) method [8]. However, there are difficulties in performing these methods. For AWE, sometimes it is difficult to obtain the derivative of the impedance matrix and the induced current vector. Singularity problem is the main weakness of the MBPE method as it needs matrix inversion to get the coefficients of the instead model. Our proposed method needs no matrix inversion and interpolates for the final results, this attribute virtually leads the approach to bandwidth unlimited interpolation with a single rational function.

The reminder of this paper is organized as follow. In Section II we describe the basic theory of the adaptive rational function sampling based on Store-Bulirsch algorithm. The numerical results which demonstrate the accuracy and efficiency of our proposed method are given out in Section III. Conclusion and discussion are provided in Section IV.

II THEORY

The interpolation function is described in the form of a fractional polynomial function with numerator of order N and denominator of order D for one parameter, most often the frequency f by

$$S(f) = \frac{a_0 + a_1 f + a_2 f^2 + \dots}{1 + b_1 f + b_2 f^2 + \dots} = \frac{a_0 + \sum_{n=1}^{N} a_n f^n}{1 + \sum_{d=1}^{D} b_d f^d} \qquad (1)$$

For given orders of N and D the coefficients a_i and b_j can be determined from $k=N+D+1$ samples of $S(f)$ by solving a set of linear equations. The Store-Bulirsch algorithm does not require the inversion of a matrix to get the coefficients of the rational function. The detail of the algorithm is outlined as follow.

Suppose that there a group of samples $(f_i, S(f_i))$, $i=1, \dots, k$ available for obtaining a rational function interpolation of function value $S(f)$ at any $f \in (f_1, f_2)$ whereas the explicit expression of rational function is unknown. A triangle table shown in Fig. 1 illustrates the recursive process of Store-Bulirsch algorithm. The recursive algorithm starts with the initial condition

$$R_{i,1} = S(f_i) \quad i = 1, \dots, k \qquad (2)$$

This constructs the first column of the triangle table. Starting from the second column, all the elements are obtained using a recursive formula associated with two or three elements in the preceding columns.

Store-Bulirsch algorithm provides two 'triangle rules'

$$R_{j,k} = \left((f - f_j)R_{j+1,k-1} + (f_{j+k} - f)R_{j,k-1}\right) \Big/ (f_{j+k} - f_j) \qquad (3)$$

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and

$$R_{j,k} = (f_{j+k} - f_j) \Big/ ((f - f_j) \Big/ R_{j+1,k-1} + (f_{j+k} - f) \Big/ R_{j,k-1}) \quad (4)$$

and one 'rhombus rule'

$$R_{j,k} = R_{j+1,k-2} + (f_{j+k} - f_j) \Big/ ((f - f_j) \Big/ (R_{j+1,k-1} - R_{j+1,k-2}) + (f_{j+k} - f) \Big/ (R_{j,k-1} - R_{j+1,k-2})) \quad (5)$$

Based on these recursive rules, our SB-AFS algorithm is organized as follow

Let the whole frequency band is defined from f_1 to f_2.The results recursively calculated by formula (3), (4), (5) are $S1(f)$, $S2(f)$, and $S3(f)$ respectively. A testing point within the frequency band of interest is f_t and a sample is f_s. The true value from EM simulation or from experiment is $EM(f_s)$.

Step 1) Setting $f_{s1}=f_1$ and $f_{s2}=f_2$, and acquiring $EM(f_{s1})$ and $EM(f_{s2})$.

Step 2) Selecting (3) and (4) as the recursive formula and finding the point at which the maximum sampling error occurs, say at point f_{s3}, such that

$$| S1(f_{s3}) - S2(f_{s3}) | = \max_i | S1(f_{ti}) - S2(f_{ti}) | \quad (6)$$

Step 3) Conducting $EM(f_{s3})$. If a given error tolerance EPS is larger than the sampling error in step 2, that is $| S1(f_{s3}) - S2(f_{s3}) | < EPS$, the process switches to step 6 for termination; otherwise, adds the sample of $\{ f_{s3}, EM(f_{s3}) \}$ into the sample collection and goes to next step for finding next samples.

Step 4) Changing the recursive formula into (3) and (5), comparing the sampling value at all testing points and finding the point at which the maximum sampling error occurs, say at point f_{s4}, such that

$$| S1(f_{s4}) - S3(f_{s4}) | = \max_i | S1(f_{ti}) - S3(f_{ti}) | \quad (7)$$

Step 5) Having had $EM(f_{s4})$, if $| S1(f_{s4}) - S3(f_{s4}) | < EPS$, which means that the samples obtained are enough for the sampling for rule (3), the process goes to step 6; otherwise, adds the point $\{ f_{s4}, EM(f_{s4}) \}$ into the sample collection and goes back to step 2 to find next samples.

Step 6) Sufficient samples have been acquired for sampling. Saving the sample collection $\{ f_{si}, EM(f_{si}), i=1,\ldots,N \}$, which is the only required knowledge for rational function interpolation, and N is the number of samples in the last step of the AFS.

After an iterative AFS process terminates, three recursive formula, i.e. (3), (4) and (5), should be sufficiently close to each other. In this study, (3) is used for interpolating frequency response at any given frequency f.

III NUMERICAL EXAMPLES

To test the accuracy and efficiency of the interpolation method described in Section II, the scattering parameter of two different FSS structures are calculated over a wide frequency band. The double-screen flat panels [9] is investigated as the first example, and testing frequency band is from 1 GHz to 30 GHz, with the incident angle $\theta = 0°$, $\Phi = 45°$ respectively for TM case. Our second structure example is the rectangular slots [10], which frequency band ranges from 100 GHz to 500 GHz, with the incident angle $\theta = 20°$,$\Phi = 90°$ respectively for TM case.

Both the directly calculated and the interpolated results are presented in Fig. 2 and Fig. 3. Excellent agreements between these two methods have been achieved for both the double-screen and rectangular slots cases. And also, the efficiency of our proposed method can be seen from Table. I.

IV CONCLUSION

In this paper, we have introduced a new frequency sampling approach based on the Store-Bulirsch algorithm for fast electromagnetic computation. Using the proposed method, the frequency domain response can be modeled by only one rational function with quite less number of sampling points of EM calculation than direct method. Since Store-Bulirsch algorithm is a recursive tabular rational interpolation, there is no matrix inverse required and the derived fitting model can be applicable in an almost unlimited frequency band. Through numerical examples, it can be observed that this method can result in both considerable CPU time saving and perfect agreement.

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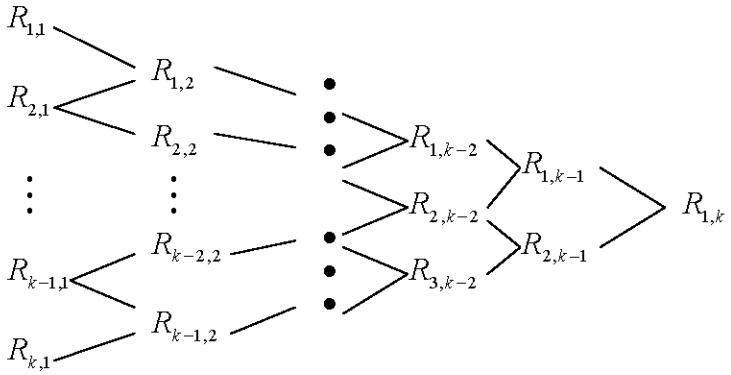


Figure 1. Tabular chart of the Store-Bulirsch algorithm for rational interpolation

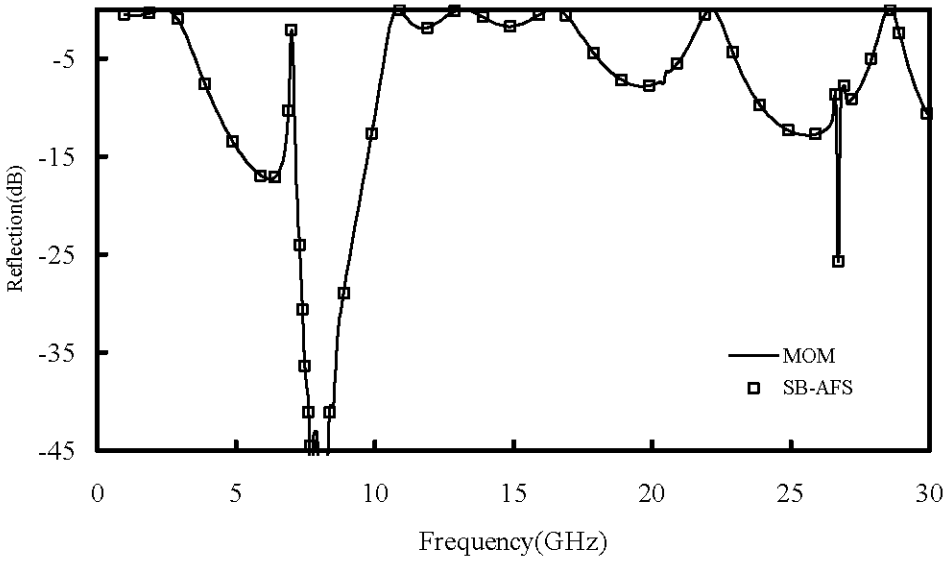


Figure 2. Reflection losses versus frequency for the double-screen FSS

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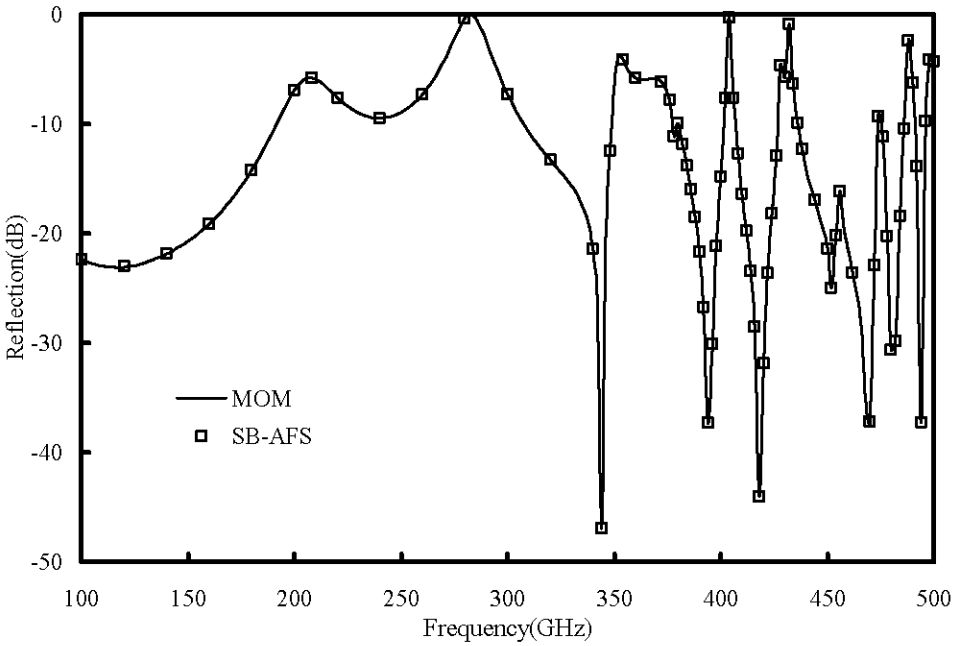


Figure 3. Reflection losses versus frequency for the rectangular slots FSS

TABLE I. Comparison of the cost and performance between direct MOM and SB-AFS method

Geometry	Calculated Nodes		Solution Time(s)	
	MOM	***SB-AFS***	***MOM***	***SB-AFS***
Double-screen	291	105	4474.45	1614.49
Rectangular slots	401	105	4483.66	1147.82

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